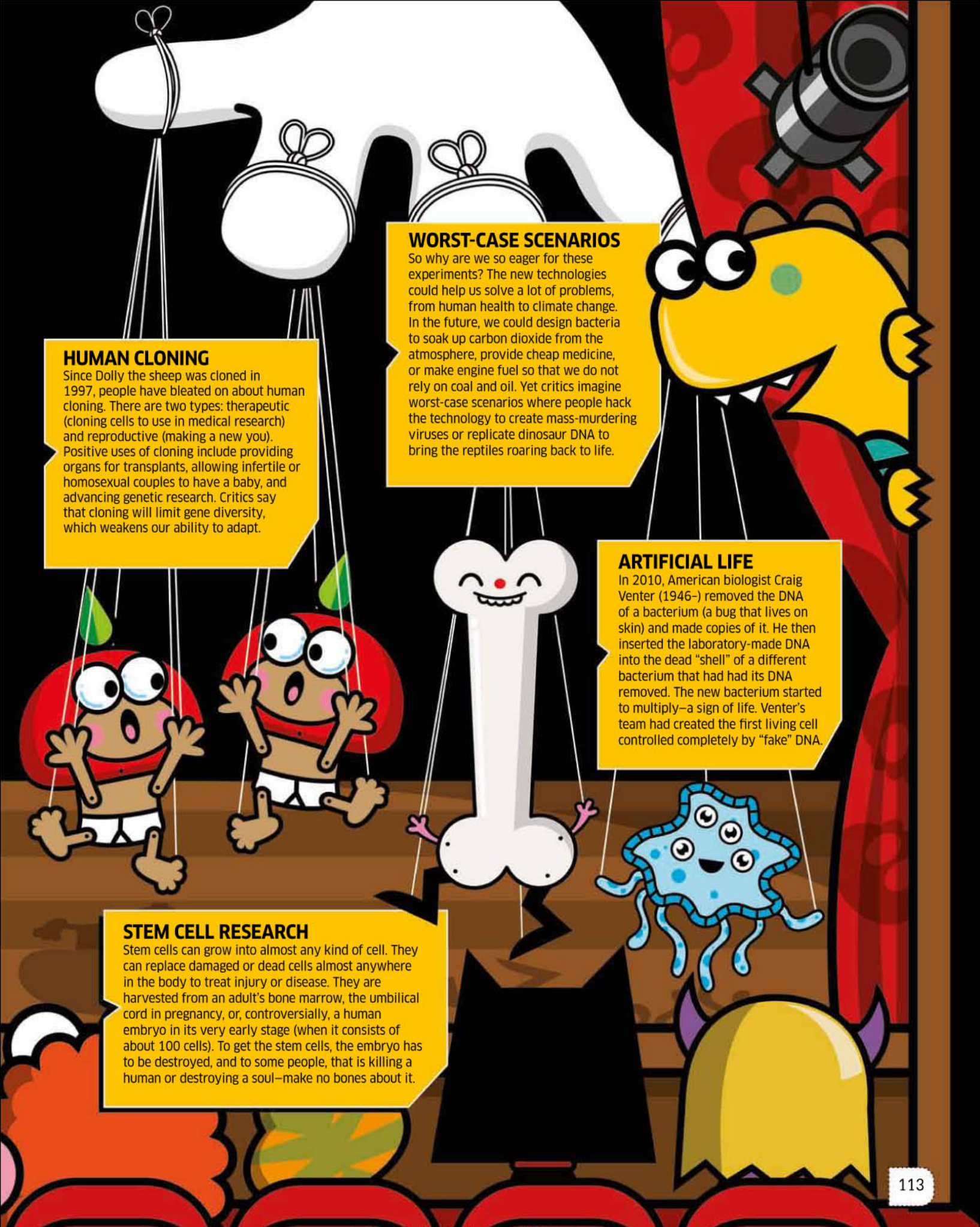




HUMAN CLONING

Since Dolly the sheep was cloned in 1997, people have bleated on about human cloning. There are two types: therapeutic (cloning cells to use in medical research) and reproductive (making a new you). Positive uses of cloning include providing organs for transplants, allowing infertile or homosexual couples to have a baby, and advancing genetic research. Critics say that cloning will limit gene diversity, which weakens our ability to adapt.

WORST-CASE SCENARIOS

So why are we so eager for these experiments? The new technologies could help us solve a lot of problems, from human health to climate change. In the future, we could design bacteria to soak up carbon dioxide from the atmosphere, provide cheap medicine, or make engine fuel so that we do not rely on coal and oil. Yet critics imagine worst-case scenarios where people hack the technology to create mass-murdering viruses or replicate dinosaur DNA to bring the reptiles roaring back to life.

ARTIFICIAL LIFE

In 2010, American biologist Craig Venter (1946-) removed the DNA of a bacterium (a bug that lives on skin) and made copies of it. He then inserted the laboratory-made DNA into the dead "shell" of a different bacterium that had had its DNA removed. The new bacterium started to multiply—a sign of life. Venter's team had created the first living cell controlled completely by "fake" DNA.

STEM CELL RESEARCH

Stem cells can grow into almost any kind of cell. They can replace damaged or dead cells almost anywhere in the body to treat injury or disease. They are harvested from an adult's bone marrow, the umbilical cord in pregnancy, or, controversially, a human embryo in its very early stage (when it consists of about 100 cells). To get the stem cells, the embryo has to be destroyed, and to some people, that is killing a human or destroying a soul—make no bones about it.